

CEPA Logic PRO — Acoustic Feature Analysis (ADS vs NORMAL)

Overview

This document presents a concise acoustic comparison between advertisement audio (**ADS**) and regular audio content (**NORMAL**) using the CEPA Logic PRO 29-feature acoustic fingerprint. The purpose of this analysis is to provide behavioural context for CEPA Logic PRO, which interprets audio conditions and adjusts system behaviour accordingly.

CEPA Logic PRO is a behavioural engine. It does not classify audio — it reacts to ADS/NORMAL signals produced by the machine-learning model used inside AdBuster PRO.

AdBuster PRO is a TV volume stabilizer. It uses a local `.pkl` model to detect ADS/NORMAL and then applies volume-control actions through IR.

Acoustic Feature Summary

CEPA Logic PRO uses a 29-feature acoustic fingerprint extracted from each audio segment:

- 13 MFCC coefficients
- 12 Chroma coefficients
- Spectral Centroid
- Spectral Bandwidth
- RMS
- Zero-Crossing Rate (ZCR)

These features describe the acoustic profile of the audio and support behavioural interpretation related to intensity, brightness, tonal structure, transient density, and compression level.

Key Acoustic Differences

Brightness (Spectral Centroid)

Advertisements contain significantly more high-frequency energy, resulting in a brighter acoustic profile.

Spectral Width (Bandwidth)

Advertisements exhibit a wider spectral spread, reflecting dense mixing and compression.

Tonal Structure (Chroma)

Chroma values are consistently lower in ADS, indicating blurred tonal structure caused by compression and layered sound effects. Normal audio shows stable, natural tonal distribution.

Transient Density (ZCR)

Advertisements contain more transients and rapid signal changes.

MFCC Profile

MFCC coefficients show strong, consistent differences across multiple bands. ADS exhibit higher energy, stronger compression artifacts, and more irregular spectral patterns. NORMAL audio remains smooth and natural.

Behavioural Interpretation (CEPA Logic PRO)

ADS — high-intensity context

- triggers protective behaviour
- activates comfort preservation
- increases sensitivity to spikes
- stabilizes transitions

NORMAL — low-intensity context

- maintains baseline behaviour
- preserves natural dynamics
- reduces intervention pressure

CEPA Logic PRO interprets acoustic context — it does not classify audio.

Verification Summary

All functional audio groups were evaluated using the machine-learning model embedded in AdBuster PRO. Each category produced stable and conflict-free results, confirming correct operation of the model and reliable behavioural signal delivery to CEPA Logic PRO.

- ADS content was consistently detected as advertisement.
- NORMAL speech and program audio were consistently interpreted as non-advertisement.
- MUSIC segments aligned with NORMAL behaviour.
- TALK segments aligned with NORMAL behaviour.

These tests confirm that the ML model operates reliably, and that AdBuster PRO and CEPA Logic PRO interact without conflicts or misinterpretation.

Final Conclusion

Advertisements and normal audio differ clearly across all major acoustic dimensions. These differences form strong and consistent behavioural signals for CEPA Logic PRO, enabling precise and context-aware reactions across all operating states. The acoustic fingerprint demonstrates stable separation between ADS and NORMAL content, ensuring reliable behavioural interpretation within the CEPA Logic PRO engine.